

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-abf0bb8c5ac9d4036.jsonl`  
- SHA-256: `03dbd50c580d5457d40119ad1263af61f9ce85f33f4ad588e56f3b45577c0a0e`  
- Source modified: `2026-06-16T03:19:35+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:04.324Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/abs/2303.07347>

****Title:**** TriDet: Temporal Action Detection with Relative Boundary Modeling (CVPR 2023)

****Found via:**** Single-stage TAL actionness + boundary heads, small data search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:06.427Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:19:20.540Z)

TriDet: Temporal Action Detection with Relative Boundary Modeling

(1) Trident-head & Relative Boundary Modeling

****Problem Addressed:**** "Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos."

****Solution:**** The paper proposes "a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary." This approach directly addresses the boundary ambiguity problem by modeling the distribution of probabilities relative to action endpoints rather than predicting fixed boundaries.

(2) SGP (Scalable Granularity Perception) Layer

The paper describes it as: "an efficient Scalable-Granularity Perception (SGP) layer to mitigate the rank loss problem of self-attention that takes place in the video features and aggregate information across different temporal granularities."

The SGP layer is designed to be ****transformer-aware**** (addressing self-attention rank loss) and handles multi-scale temporal information aggregation. It's characterized as "efficient" but the content doesn't explicitly confirm whether it's convolution-based or transformer-free.

(3) Reported Performance Metrics

****THUMOS14:**** "TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%"

****HACS & EPIC-KITCHEN 100:**** Mentioned as benchmarks where state-of-the-art is achieved, but specific numbers aren't provided in the abstract.

(4) Inference Speed/Efficiency

"with lower computational costs, compared to previous methods. For example, TriDet hits an average mAP of 69.3% on THUMOS14... but with only 74.6% of its latency."

(5) Publication Details

****Venue & Date:**** CVPR 2023

****Submission:**** "Submitted on 13 Mar 2023 ([v1](https://arxiv.org/abs/2303.07347v1)), last revised 16 Mar 2023 (this version, v2)"

****Authors:**** Dingfeng Shi, Yujie Zhong, Qiong Cao, Lin Ma, Jia Li, Dacheng Tao

4. assistant (2026-06-16T03:19:22.958Z)

This is the primary arXiv abstract page for the TriDet paper (CVPR 2023). I have enough to extract falsifiable claims.

5. user (2026-06-16T03:19:30.514Z)

Structured output provided successfully

6. assistant (2026-06-16T03:19:35.807Z)

Extracted claims from the TriDet (CVPR 2023) primary arXiv source. The source is a primary research paper directly relevant to Dimension 1 (windowing/boundaries) of the research question: its Trident-head relative-boundary modeling and SGP layer are concrete recent boundary-localization techniques, though their direct transfer to a tiny-data 1-D trajectory setting (vs. THUMOS14-scale benchmarks) remains the open question to flag.